

1. Administrative & Study Identification

Full Study Title: Study of Patient With Frequent COPD Exacerbations **Short Title/Acronym:** CERERA Study **Protocol Number:** Internal Study ID - CERERA
NCT Number: NCT07221981 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Non-interventional, observational study) **Phase of Development:** Observational / Real-World Evidence **Study Status:** Recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the characteristics, management, and exacerbation rates of patients with frequent COPD exacerbations in a real-world, clinical setting. **Secondary Objectives:** To describe the frequency and severity of exacerbations, longitudinal treatment patterns, and healthcare resource utilization among this patient population.

2.2 Background & Rationale To address the evidence gap in the real-world clinical profiles and disease management of patients experiencing frequent COPD exacerbations. Understanding how these patients are managed in routine clinical practice is essential to identifying unmet needs, improving targeted interventions, and reducing the significant morbidity and healthcare burden associated with recurrent COPD exacerbations.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Ambispective: retrospective-prospective) **Control Type:** Single-arm (Cohort) **Blinding:** Open-label (Observational) **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 2,000 participants
Number of Sites: Multicenter **Estimated Study Start:** 2024
Estimated Primary Completion: 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of inclusion. 2. Confirmed diagnosis of Chronic Obstructive Pulmonary Disease (COPD). 3. Documented history of frequent COPD exacerbations in the preceding observation period.

4.2 Exclusion Criteria 1. Current or previous diagnosis of clinically important pulmonary disease other than COPD (e.g., asthma). 2. Concurrent participation in any interventional investigational drug trial. 3. Life expectancy of less than 12 months or inability to comply with data collection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Standard of care as prescribed by the patient's treating physician). **Route of Administration:** N/A (Routine clinical care). **Duration of Treatment:** Observation period spanning both retrospective medical history and prospective follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: All standard, routine maintenance therapies for COPD and comorbidities. **Prohibited:** Investigational medications administered as part of a concurrent interventional clinical trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures		
1	Screening/Baseline	Day 0	Informed Consent, Chart Review, Retrospective Data Extraction	Interim
2	Standard of care	Routine clinical visits, Prospective Data Collection (Exacerbations, EHR review)	End of Study	End of Follow-Up
		Final Data Abstraction, Exacerbation Rate Calculation, Healthcare Resource Utilization logging		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized rate of moderate or severe COPD exacerbations over the study observation period. **Measurement Method:** Electronic Health Record (EHR) data abstraction and patient-reported clinical events during standard care visits.

7.2 Secondary & Exploratory Endpoints * Time to the first moderate or severe COPD exacerbation during the prospective period. * Healthcare resource utilization (e.g., hospitalization frequency, ER visits) attributed to COPD exacerbations.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard pharmacovigilance reporting as required for observational studies. Data is extracted retrospectively and prospectively from medical charts. **Serious Adverse Events (SAEs):** Routine reporting within 24 hours

if specifically associated with a sponsor's prescribed product under standard safety reporting regulations. **Data Monitoring Committee (DMC):** Internal sponsor review board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Observational Cohort / Full Analysis Set (All enrolled, eligible patients). **Sample Size Justification:** A target $N = 2,000$ ensures sufficient statistical power to precisely estimate the exacerbation rate and descriptive statistics for disease subpopulations. **Handling of Missing Data:** Data will be analyzed as observed with no formal imputation, consistent with standard real-world evidence methodologies.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), applicable Good Clinical Practice (GCP) guidelines, and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring; periodic data quality checks and source data verification of the electronic Case Report Forms (eCRF) against medical records. **Audit Trail:** Validated Electronic Data Capture (EDC) system with full audit trails and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CERERA **Registry Number:** NCT07221981 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Study of Patient With Frequent COPD Exacerbations **Official Regulatory Title:** A multicenter, non-interventional, observational, ambispective (retrospective-prospective) study of Patients with Frequent COPD Exacerbations

12. Clinical Framework (COPD Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD) **Diagnosis Threshold:** Confirmed COPD diagnosis with documented frequent exacerbations **Medical Methodology:** Standard of Care / Real-world maintenance therapy **Optimization Target:** Reduction of COPD exacerbation rate and healthcare resource utilization

13. Study Procedures & Methodology

Management Pathway: Standard real-world clinical pathway for COPD management **Education Protocol (HCP):** N/A (Non-interventional observational design) **Education Protocol (Patient):** Routine

disease education per clinical practice **Quality Audit Mechanism:** eCRF monitoring, source data verification, and completeness tracking for EHR extraction

14. Observation & Follow-Up Schedule

Total Duration: Full ambispective period (retrospective chart history + prospective follow-up) **Onsite Visit Cadence:** Aligned strictly with routine standard-of-care clinical visits **Remote Monitoring Frequency:** Periodic EHR data abstraction intervals **Checklist Review Interval:** Baseline, interim data cuts, and end of prospective observation

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				